ME 3060-004 Machine Design Final Project: Gantry Crane

Ashley Burgess, William Meany, Peter Miller

Department of Mechanical Engineering, Clemson University, Clemson, SC, 29634

04 December 2025





PAGE 2: TABLE OF CONTENTS

1: Introduction **4**
1.1 Case Description 4
1.2 Assumptions 5
2. Design Overview **5**
2.1 Material Selection Process 5
2.2 Factor of Safety 6
2.3 Overall Sketch and Free-body Diagram 7
3. Horizontal Beams **7**
3.1 Description and Requirements 7
3.2 Assumptions 8
3.3 Sketch and Free-Body Diagram 8
3.4 Structural Analysis and Failure Criteria 9
4. Crossbar **11**
4.1 Description and Requirements 11
4.2 Assumptions 11
4.3 Sketch and Free-Body Diagram 12
4.4 Structural Analysis and Failure Criteria 12
5. Upright Column **13**
5.1 Description and Requirements 13
5.2 Assumptions 14
5.3 Sketch and Free-Body Diagram 14
5.4 Structural Analysis and Failure Criteria 15
6. Pulley Bracket **17**
6.1 Description and Requirements 17
6.2 Assumptions 17
6.3 Sketch and Free-Body Diagram 17
6.4 Structural Analysis and Failure Criteria 17
7. Pressure Vessel (of Hydraulic Piston) **17**
7.1 Description and Requirements 17
7.2 Assumptions 17
7.3 Sketch and Free-Body Diagram 18
7.4 Structural Analysis and Failure Criteria 18
8. Column Base (w/ Bolted Connection) **20**
8.1 Description and Requirements 20
8.2 Assumptions 20
8.3 Sketch and Free-Body Diagram 21
8.4 Structural Analysis and Failure Criteria 21

8.5 Finite Element Analysis 23
9. Piston **23**
9.1 Description and Requirements 23
9.2 Assumptions 24
9.3 Sketch and Free-Body Diagram 25
9.4 Column Analysis 26
10. Pulley **28**
10.1 Description and Requirements 28
10.2 Assumptions 28
10.3 Sketch and Free-Body Diagram 28
10.4 Structural Analysis and Failure Criteria 28
11. Crane Hook **29**
11.1 Description and Requirements 29
11.2 Assumptions 29
11.3 Sketch and Free-Body Diagram 30
11.4 Structural Analysis and Failure Criteria 30
11.5 Fatigue Analysis 32
12. Conclusion **34**
13. References **35**

1: Introduction

1.1 Case Description

This report discloses the design and developmental process gantry crane intended for maritime usage. These cranes are used in marines to lift boats out of the water and transport them onto a trailer or into storage. Below, figure 1.1 depicts a gantry crane handling a boat.

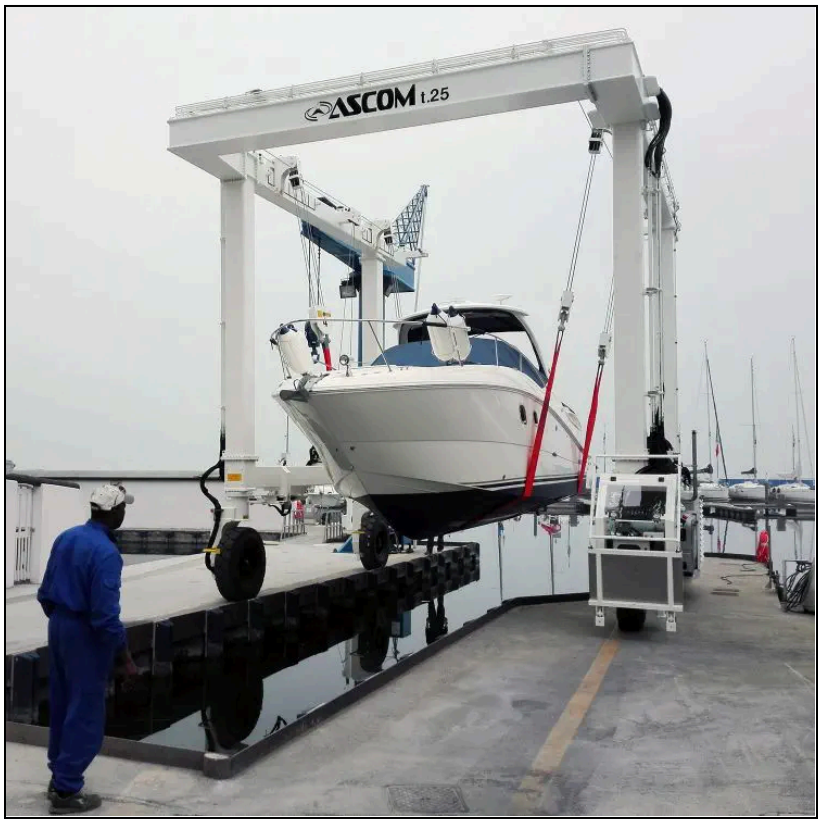


Figure 1.1: Gantry Crane Operating in Marina

Gantry cranes are a valuable piece of equipment for these marinas, and are often the most efficient way to remove a boat from the water. Occasionally, boats need to be removed from the water for inspection, or repairs. Small recreational boats are able to be transported with a trailer, but larger boats, ones that a 25 ton gantry crane would be built to handle, could be too large for frequent trailering. This is where the gantry crane is important. The ability to quickly remove boats from the water, and not requiring much space to operate are the two largest advantages. Due to its importance, these cranes need to operate reliably, and repeatedly. This report will highlight the design process that this team followed to design a gantry crane.

1.2 Assumptions

The following are general assumptions that were made pertaining to the general use of the completed product. The gantry boat hoist detailed in this report was designed assuming a customer/ user base of recreational boat owners with a competent familiarity with appliances akin to the disclosed hoist. It is assumed that this hoist will lift boats of the Class A or Class 1 federal classifications. These boats may weigh anywhere from 850 lbs (i.e. personal watercrafts) to 8,700lbs (for cabin cruisers or speed boats). This boat hoist is dimensioned to comfortably lift Class 1 watercrafts that adhere to these average weight loads. The pressure vessel is designed to meet the criteria disclosed; namely, it shall be capable of a working pressure necessary to translate the pulley bracket loaded at 25 tons. It is assumed that this hoist will be installed in a public medium-sized marina. The gantry crane will have four support gussets that will be bolted to a set of wheels that will facilitate the crane's movement along a loading ramp. It is assumed that the hoist will be used an average of once per day accounting for the on and off seasons of variable uses. It is further assumed that the high use seasons will align with the summer time and high maintenance seasons. It is further assumed that all materials are flawless and high quality from the foundry. The analysis is only valid if the materials have constant properties that perform according to the tabulated values. All loads are assumed to be static, though this report recognizes that the boat will be moving during the use of the crane. Static loads are valid assumptions as the process demands a design based on failure. It is further assumed that the two horizontal sidebars are perfectly symmetrical.

Along with this, the load on the pulleys is assumed to be strictly vertical, as the latches that will connect the boat to the hooks are assumed to be directly under the pulley. This is assumed for all parts except the crossbar (Section 4) and the column base (Section 8). This is because the main objective of these components is to be a reaction force that prevents the system from collapsing in on itself due to the torsion and moments it would cause.

Because there are only two hooks, one on each side, in real-life circumstances, it is physically impossible to lift the boat without it tipping over. Because of this, we will assume that the center of mass/gravity of the boat will act in the same plane as the pulleys and hooks.

Along with this, the pulleys must be able to reach the center of mass of the boat. The rod and pressure vessel are designed to be able to translate back and forth along the horizontal beams. It is assumed that the interval 10 ft (as will be seen in Section 7 of the Pressure Vessel

and Section 9 of the Rod/Piston) will cover the entire range of where the center of mass of the boat could be.

2. Design Overview

2.1 Material Selection Process

There are many factors that need to be considered when selecting a single material for the gantry crane design. Both structural integrity as well as long-term durability must be considered in order to design an effective and safe gantry crane. These two factors manifest themselves in two different criteria that we mainly considered: the yield strength of the material and its ability to withstand environmental impacts.

Being near the water and in outdoor conditions, it is vital that the material chosen is as corrosion-resistant as possible. Sprays and splashes of water, humidity, wind, oxidation, and temperature fluctuation are all potential causes of corrosion within the gantry crane. These will diminish the durability and safety of the crane, as corrosion can lead to cracks and structural failure within the material.

The gantry crane must also be able to lift 25 tons, which is quite a feat. Thus, the material must be able to withstand the extreme stresses that the load causes. Since the design only uses two pulleys, the force will be concentrated as two point loads, as opposed to being distributed over an area. This causes larger moments and stresses along the material.

Finding a balance between these two factors (namely, strength and durability) is extremely difficult, but with ASTM A500 Steel (Grade C), it is manageable. This material was chosen in large part due to its relatively high yield strength of 50 [kpsi] and its ready availability from vendors in rectangular beam form. Although the corrosion resistance of the material is subpar, steps can be taken to improve its durability. Many types of coatings can be applied to the material to make it more suitable for the harsh, outdoor environment. Epoxy paint coatings and powder coatings are two excellent examples of high-performance coatings that have the capabilities to protect against corrosion.

Although ASTM A500 is the primary material we are using, some components of the crane will be made of other materials. This is because some parts are too small, or the highly specific requirements necessitate the use of another better-suited material. For example, the crane hook is to be designed with 1040 CD Steel (yield strength of 71 kpsi). This part has very strict

guidelines, and it is not ideal to use 304 Stainless Steel because of its lower yield strength. Although 1040 CD Steel has poor corrosion resistance, the component is small, cheap, and easy to replace, justifying the decision to change materials despite the outdoor conditions.

2.2 Factor of Safety

When designing any type of structural or mechanical system, a Factor of Safety (FoS) is applied to make sure that a component or system can withstand loads larger than expected. This acts as a margin of safety, or buffer, with the maximum stress that a component or system can withstand, so that in the event of unexpected load or failure, the material will stand strong.

This gantry crane has been designed as a hoisting device, meaning that the system is designed only to raise and lower a boat in the vertical direction. As per federal requirements in the United States, hoisting devices like this must be designed to have a factor of safety between 2:1 and 3:1. Due to these requirements, it is assumed that there will only be vertical forces on the lift. Further, given how small the tolerances are, a conservative factor of safety of 2:1 was chosen for the system. This allows for plenty of leniency for any excess loads and stresses, but also more design freedom as the reporters can design for a higher allowable stress.

2.3 Overall Sketch and Free-body Diagram

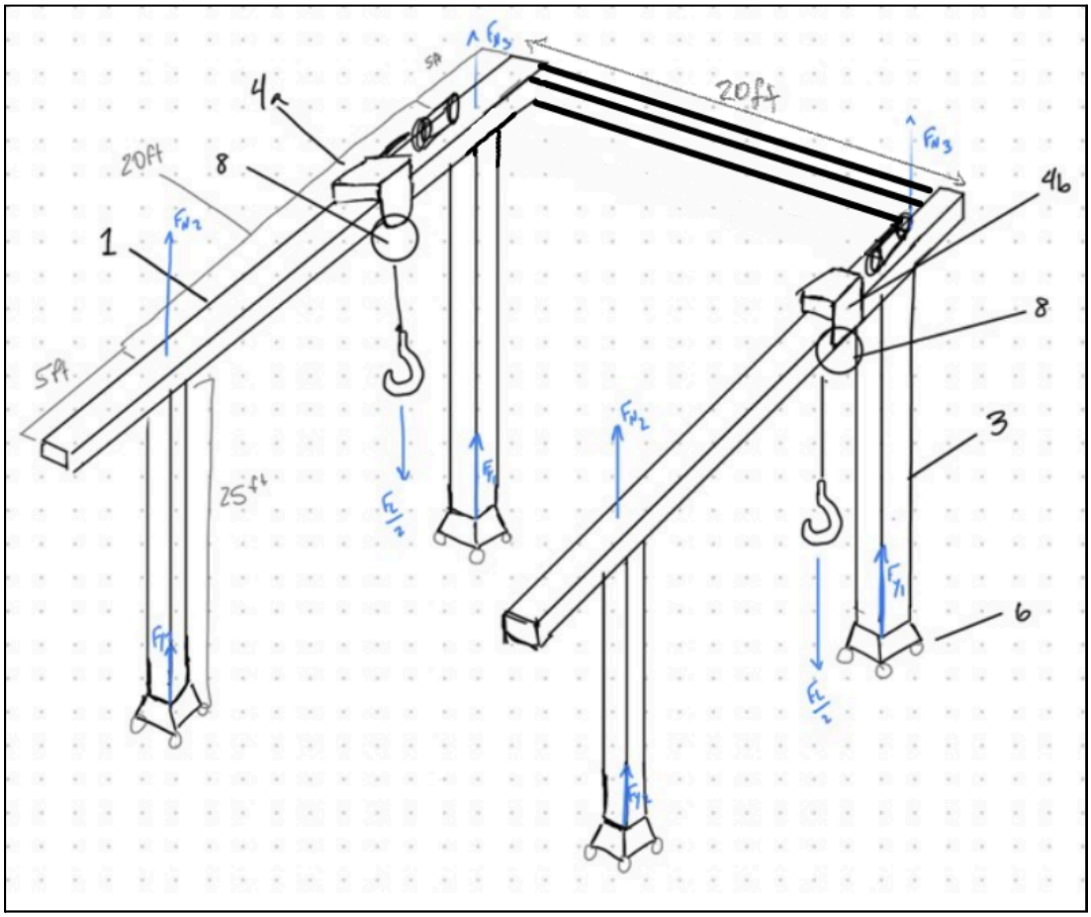


Figure 2.3: Overall Free-body diagram

3. Horizontal Beams

3.1 Description and Requirements

The two horizontal beams are one of the heavier and larger components of the gantry crane design. Sitting on top of the upright columns, these beams are meant to support half the load of the boat, 12.5 tons each, to ultimately be able to support the stress from the lifting force. Along with this, these beams are what support and guide the pulley brackets and pistons forward and backward.

The horizontal side beams have many requirements that severely restrict the suitable choice of material, size, geometry, and the location upon which to support these horizontal beams with the upright columns. These explicit requirements include: first, the beams have to be a rectangular tube with rounded corners; second, the beams must support the maximum load, and

finally, the size must be available for purchase from a metal supplier. These, as well as other restrictions, such as the maximum boat length of 50 feet, and the limited flexibility, further requires specific material selection.

Under all these requirements, the horizontal beam chosen was from *Metals Depot*. This beam has a base $b = 8\,in$, height $h = 12\,in$, and thickness $t = 0.5\,in$. The beam length $l = 30\,ft$, and although the website does not have lengths this long, *Metals Depot* is able to custom make beams for different lengths for a higher price.

3.2 Assumptions

The major assumptions that are made for the horizontal beams are similar to those made for most other components. The load is assumed to be static. Even though the crane is designed to move the boat forward and backward using the pulley bracket, it is valid to assume that this force is still static, due to the nature of the failure calculations.

It is also valid to assume that the weight of the beams themselves are negligible. Although the weight of these beams are decently heavy, they have a negligible effect on the overall stress that is caused by the 25-ton weight of the boat.

The last and main assumption is that the force from the pulley is strictly vertical. This leads to absolutely no torsional force along the beam. Because of this, the failure calculations only need to be focused on the failure that can occur from the bending stress σ_{bend}.

3.3 Sketch and Free-Body Diagram

The following figure 3.3 is a sketch of one of the symmetrical horizontal side bars, its respective shear-moment diagram plotted against distance, and its cross-sectional geometry and area.

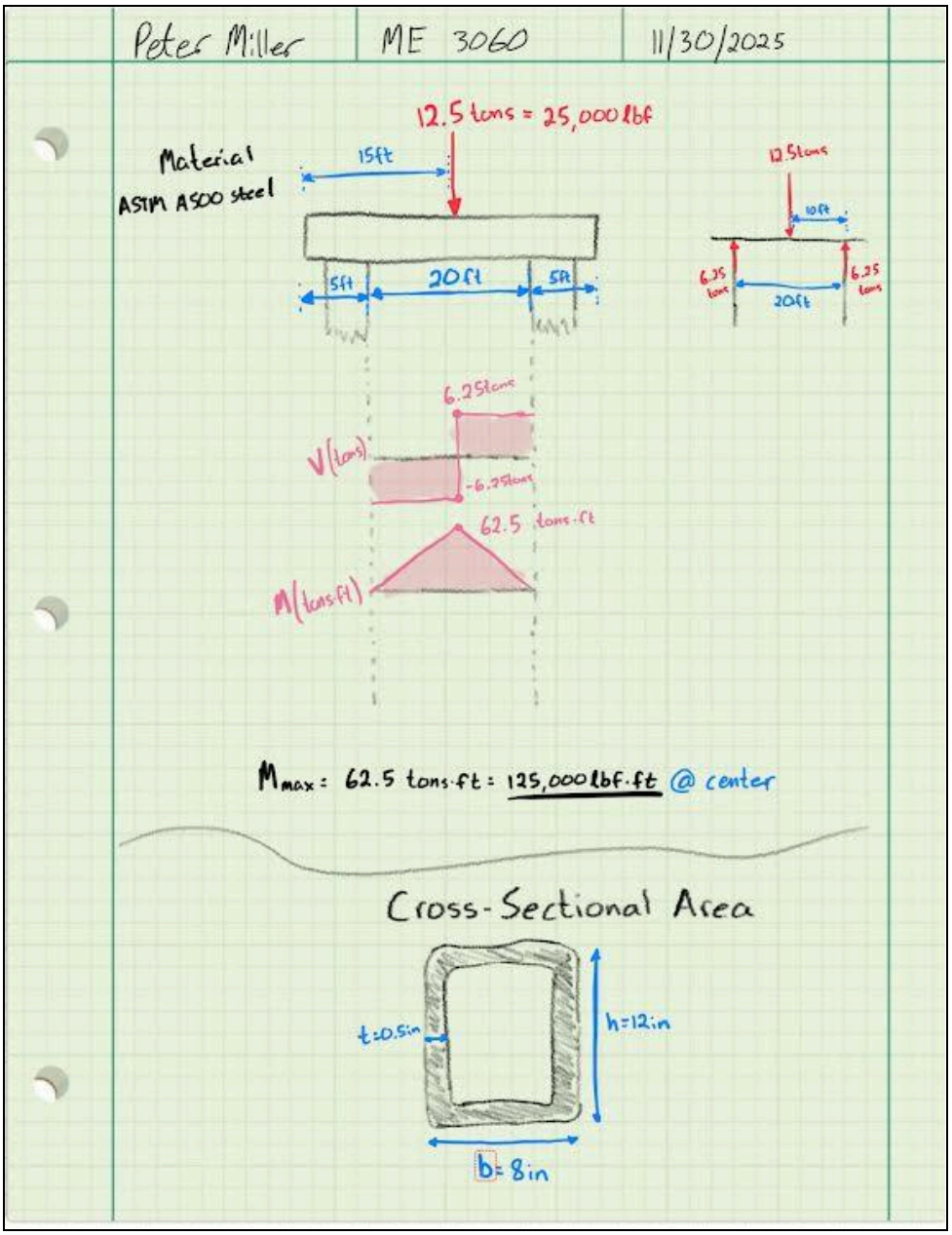


Figure 3.1: Sketch and FBD of symmetrical side horizontal beams (denoted as part 1).

3.4 Structural Analysis and Failure Criteria

To meet the failure criteria for the horizontal beams, the bending stress must be less than the allowable stress. We start by finding the allowable stress, σ_{allow}, using the ratio of the yield stress of the material (ASTM A500) and the Factor of Safety.

$$\sigma_{allow} = \frac{\sigma_y}{FoS} \tag{3.1}$$

$$\sigma_{bend} = \frac{Mc}{I} \tag{3.2}$$

To find the bending stress in the beam (3.2), we must first find the maximum moment M, the moment of inertia I, and the centroidal axis c.

$$M_{max} = \frac{PL}{4} \tag{3.3}$$

Using the maximum moment equation (3.3) is the first step to finding the bending stress. The maximum moment will occur at the distance furthest from both supporting columns. This is in the very center of the beam, where the distance from both columns is the same (10 ft).

$$I = \frac{bh^3 - (b-2t)(h-2t)^3}{12} \tag{3.4}$$

Finding the moment of inertia is the next step in finding the bending stress. The moment of inertia for a hollow rectangular beam is given above (3.4).

$$c = \frac{h}{2} \tag{3.5}$$

The centroidal axis is then found (3.5), and the bending stress σ_{bend} can be calculated from equation 3.2. If the bending stress σ_{bend} is less than the allowed stress σ_{allow}, then our part will not fail. This can be checked by finding the actual Factor of Safety below.

$$FoS_{real} = \frac{\sigma_{allow}}{\sigma_{bend}} \tag{3.6}$$

To meet the failure criteria for the horizontal beams, the bending stress must be less than the allowable stress. We start by finding the allowable stress, using the ratio of the yield stress of the material (ASTM A500) and the Factor of Safety.

$$\sigma_{allow} = \frac{50{,}000\ psi}{2} = 25\ kpsi \tag{3.1}$$

Maximum load $P = 25,000\ lbf$ and the Gap length of $L = 20\ ft$.

$$M_{max} = \frac{(25{,}000\ lbf)(20\ ft)}{4} = 125,000\ lbf * ft \tag{3.2}$$

This coincides with the shear and moment diagram found in figure 3.1 above.

Then, we chose a specific beam from *Metals Depot* as per the requirement of selecting a rectangular tube with rounded corners, with the dimensions of $b = 8\,in$, $h = 12\,in$, and $t = 0.5\,in$.

$$I = \frac{(8in)(12in)^3 - (8in - 2(0.5in))(12in - 2(0.5in))^3}{12} = 375.58\,in^4 \quad (3.3)$$

$$c = \frac{12in}{2} = 6\,in \quad (3.4)$$

$$\sigma_{bend} = \frac{Mc}{I} = 23.96\,kpsi \quad (3.5)$$

The bending stress $\sigma_{bend} = 23.96\,kpsi < \sigma_{allow} = 25\,kpsi$, meeting the failure criteria. This is checked by finding the actual Factor of Safety.

$$FoS_{real} = \frac{25\,kpsi}{23.96\,kpsi} = 2.09 \quad (3.6)$$

4. Crossbar

4.1 Description and Requirements

The crossbar is a horizontal member connecting both sides of the gantry crane. It serves a few important roles. The first role is to connect the left and right horizontal beams, and is the only point of connection between them. Second, it counteracts the bending moment when lifting a more central weight. This allows the rest of the crane to keep its shape, and ensures a compound failure does not occur. The crossbar must be designed with this importance in mind.

4.2 Assumptions

When designing the column and the horizontal beams, the load was assumed to be purely vertical. When designing the crossbar, this can not be the case. A purely vertical load would leave the crossbar with zero forces acting in its plane. The assumption is that the forces will occasionally act directed into the center of the crane on occasion, and the crossbar must be properly designed accordingly.

4.3 Sketch and Free-Body Diagram

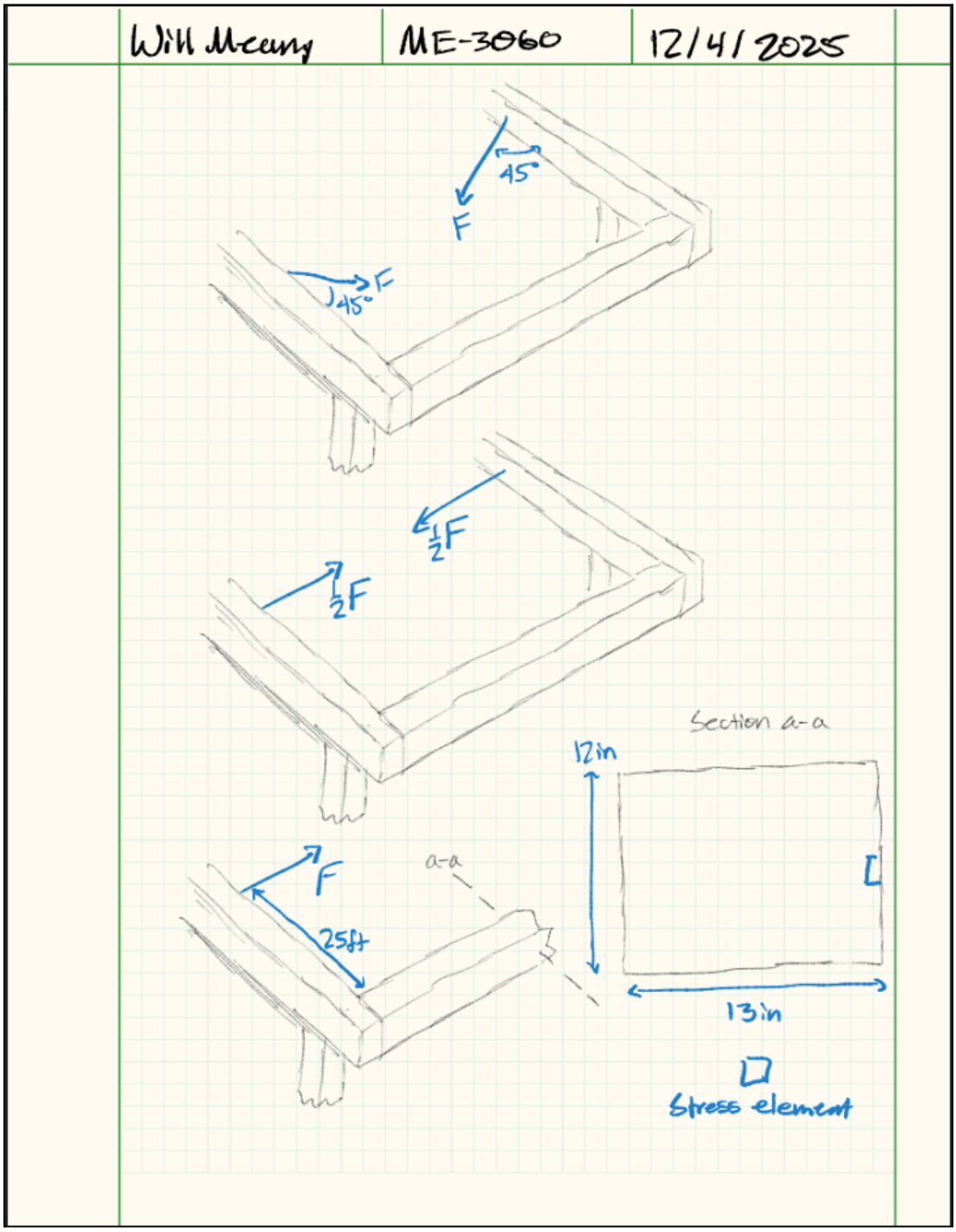


Figure 4.1: Sketch and Free Body Diagram of Crossbar

4.4 Structural Analysis and Failure Criteria

To analyze the beam, the force of lifting the boat is to be assumed at a 45 degree angle. This is a worst case scenario in which a heavy and shallow boat is being lifted. This divides the

force into two. The force in the z direction will be acting vertically through the columns, and the horizontal force will act on the crossbar. This means that half of the force of the boat will be acting on the crossbar attempting to bend it. The first step is to find the moment of inertia. The shape of the beam has had to be altered from the generic beam used for the column and horizontal supports due to the need for a larger moment of inertia about the z axis.

$$I_y = \frac{H \cdot B^3}{12} \tag{4.1}$$

The new shape of the beam having a higher height and not being hollow allows for a higher moment of inertia. The rest of the beams are loaded vertically, therefore shaped to resist vertical loads. This beam requires additional base width, as well as maintaining the same height as the horizontal beam to interface with them. To see if the beam will fail, the bending force will be solved for.

$$\sigma_{bend} = \frac{Mc}{I} \tag{4.2}$$

Finally, the factor of safety will be used to ensure the crane does not reach the yield strength of the steel.

$$\sigma_{allow} = \frac{\sigma_y}{n} \tag{4.3}$$

Plugging in the chosen values for height and base into Equation (4.1),

$$I_y = \frac{12 \cdot 13^3}{12} = 2197 in^4 \tag{4.1}$$

Next, solving for the bending stress using Equation (4.2). The variable c must be found which is the distance from the point of interest to the outside radius. This is 6.5in as seen in Figure 4.1.

$$\sigma_{bend} = \frac{25ft \cdot \frac{12in}{1ft} \cdot 25000lbf \cdot 4in}{2197in^4} = 22189.35psi \tag{4.2}$$

$$\sigma_{allow} = \frac{46000psi}{2} = 23000psi \tag{4.3}$$

The bending stress is lower than the allowable stress, therefore the beam is sized properly to resist the bending stress of the nonvertical loads

5. Upright Column

5.1 Description and Requirements

On the gantry crane, there are four upright columns that support the weight of the boat. At a maximum of 50,000lb, the boat is capable of exerting that force over no less than two columns. Due to the ability to adjust where the boat is picked up, the maximum load on a column

would be when the force is acting directly along it. This means that the weight of the boat is split in two and divided over two columns. There is no height requirement but a height of 25ft was chosen to give clearance for boats of all sizes, and allow the boat to be hoisted out of the water. Now that the columns have been described and the requirements covered the assumptions will be shown.

5.2 Assumptions

In the designing of the columns for the gantry cane, it has been assumed that the force is acting directly down through the column. This allows for a more straightforward design and cohesion of the other members involved in this design.

5.3 Sketch and Free-Body Diagram

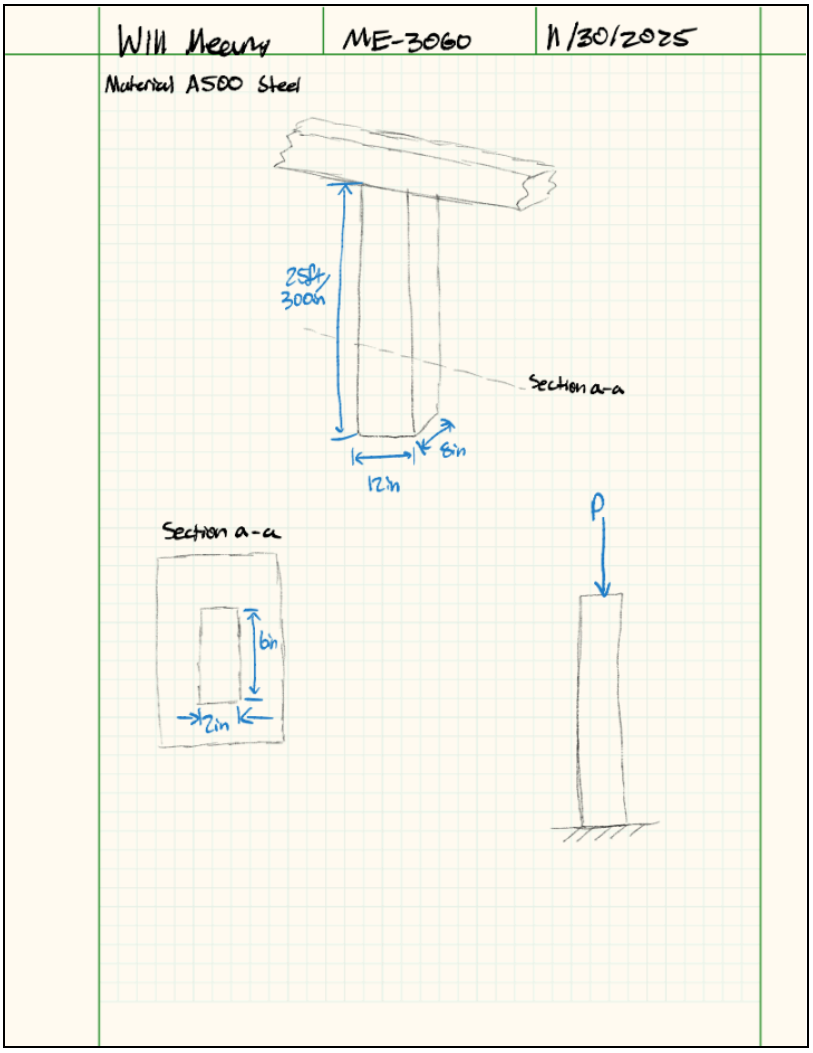


Figure 5.1: Sketch and FBD of Upright Column

5.4 Structural Analysis and Failure Criteria

The vertical supports are modeled as columns in the context of this project. This means that the first priority is to solve for buckling. The first step is to find out the categorization of the beam, due to long, intermediate, and short beams all being treated differently. Similar to the majority of members in the crane, these columns will be made from ASTM A500. In order to properly interface with the horizontal supports, these will be made to have exterior dimensions of 12inx8in, matching the horizontal beams. Even though smaller members could be used, the same size as the horizontal beams were chosen to potentially reduce complexity, and lower cost of material acquisition. To begin the column calculations, the radius of gyration (k) needs to be solved for. This is done by first finding the area (A), and then the moment of inertia (I).

$$A = (W \cdot H) - (w \cdot h) \tag{5.1}$$

$$I = \frac{(W \cdot H^3) - (w \cdot h^3)}{12} \tag{5.2}$$

$$k = \sqrt{\frac{I}{A}} \tag{5.3}$$

Next, the $\frac{l}{k}$ ratio is found, which will determine which classification the column will fall into.

$$\frac{l}{k} = \frac{height}{radius\ of\ gyration} \tag{5.4}$$

The factor of safety, is then multiplied into the critical load (P_{cr}) in order to ensure protection from failure.

$$P_{cr} = P \cdot n \tag{5.5}$$

The column is then tested for its categorization starting with long columns. If the $\frac{l}{k}$ is greater than the result of the equation, then Euler's equations for long columns apply. The constant C is representative of the end conditions of the column, and will be assumed to be 1 in this circumstance due to the difficulty to fully secure columns to a moving platform.

$$\left(\frac{l}{k}\right)_l = \left(\frac{2 \cdot \pi^2 \cdot C \cdot E}{S_y}\right)^{1/2} \tag{5.6}$$

If the column is categorized as long, then the critical load is solved for, this is the load where the beam will buckle, which is assumed to be before the yield strength on long columns. This will be

half of the actual stress on the column due to the factor of safety of two. If a column is not categorized as long, intermediate will be tested.

$$\left(\frac{l}{k}\right)_i = 0.282 * \sqrt{\frac{A*E}{P}} \tag{5.7}$$

$$\frac{Pcr}{A} = S_y - \left(\frac{S_y\, l}{2\pi k}\right)^2 \frac{1}{CE} \tag{5.8}$$

$$\sigma = \frac{P}{A} \tag{5.9}$$

$$\frac{P_{cr}}{A} = 2\sigma \tag{5.10}$$

$$n = \frac{\frac{Pcr}{A}}{\sigma} \tag{5.11}$$

Using the dimensions for the column of 12inx8in, and 6inx2in for W,B,w,b respectively, and a length of 300in:

$$A = (12in \cdot 8in) - (11in \cdot 7in) = 19in^2 \tag{5.1}$$

$$I = \frac{(12in \cdot 8in^3) - (11in \cdot 7in^3)}{12} = 197.58in^4 \tag{5.2}$$

$$k = \sqrt{\frac{197.58in^4}{19in^2}} = 3.22in \tag{5.3}$$

$$\frac{l}{k} = \frac{300in}{3.22in} = 93.03 \tag{5.4}$$

$$P_{cr} = 25,000lbf * 2 = 50,000lbf \tag{5.5}$$

$$\left(\frac{l}{k}\right)_l = \left(\frac{2 \cdot \pi^2 \cdot 1 \cdot 29,000,000psi}{46000psi}\right)^{1/2} = 111.55 \tag{5.6}$$

Since $\left(\frac{l}{k}\right) > \left(\frac{l}{k}\right)_l$ the long column assumption does not hold, and the intermediate assumption will be tested.

$$\left(\frac{l}{k}\right)_i = 0.282 * \sqrt{\frac{19in^2 * 29,000,000psi}{25,000lbf}} = 41.86 \tag{5.7}$$

Since $\left(\frac{l}{k}\right)_l > \left(\frac{l}{k}\right) > \left(\frac{l}{k}\right)_i$ this is an intermediate column, and will be solved using the intermediate equations.

$$\frac{Pcr}{A} = 46000psi - \left(\frac{46000psi * 25ft * \frac{12in}{1ft}}{2\pi 3.22in}\right)^2 \frac{1}{1*29{,}000{,}000psi} = 30004.28psi \quad (5.8)$$

$$\sigma = \frac{25000}{19in^2} = 1315.79psi \quad (5.9)$$

30,004.28psi > 2631.85 (5.10)

$$n = \frac{\frac{Pcr}{A}}{\sigma} = 22.80 \quad (5.11)$$

Therefore the column will support the load, and not buckle under the weight. They more than exceed the factor of safety, but this is due to them matching the same dimensions as the horizontal beams.

6. Pulley Bracket

6.1 Description and Requirements

6.2 Assumptions

6.3 Sketch and Free-Body Diagram

6.4 Structural Analysis and Failure Criteria

7. Pressure Vessel (of Hydraulic Piston)

7.1 Description and Requirements

The pressure vessel (hydraulic piston) is designed to translate the pulley bracket forward and backward along the horizontal beams. The main requirement of the vessel is the ability to translate the bracket while supporting the maximum load of 25 tons using its internal pressure. It must be able to interface with the rod, and be able to fit on the horizontal beams. The pressure vessel ensures that the hooks are always located at the center of mass.

7.2 Assumptions

The assumption made for the pressure vessel is that the center of mass of the boat or object being lifted, resides in the range of motion of the hydraulic piston.

7.3 Sketch and Free-Body Diagram

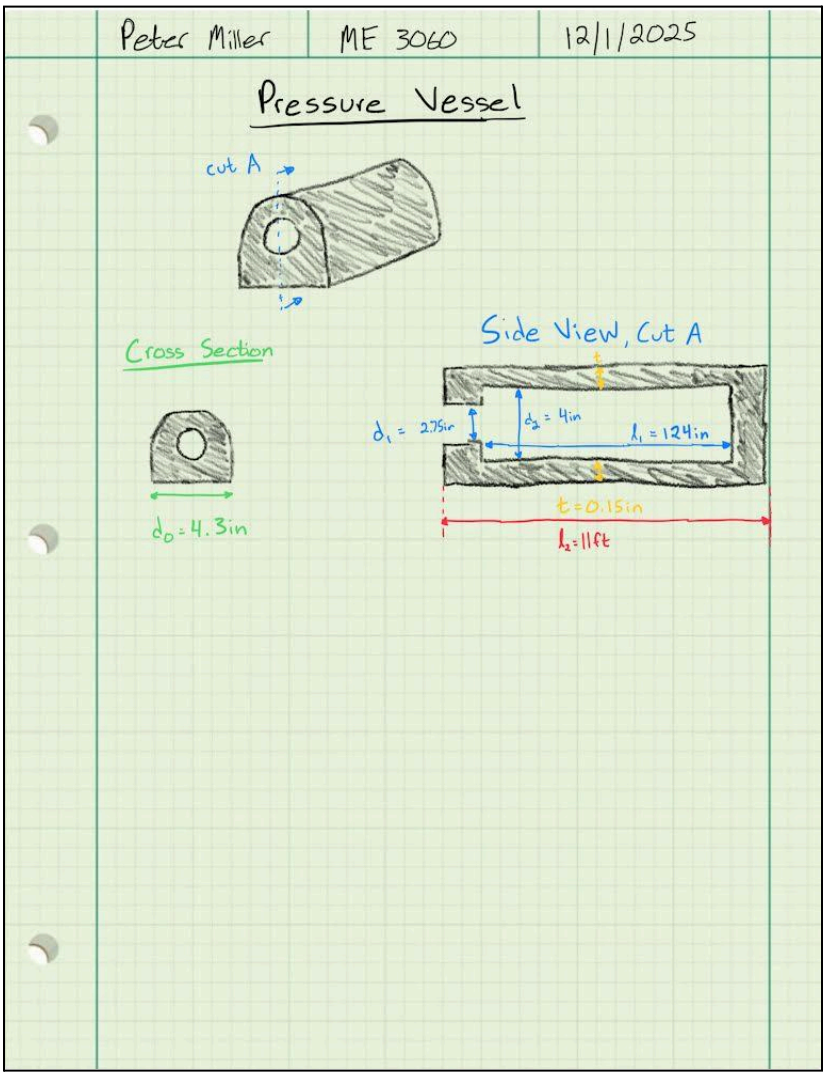


Figure 7.1: Sketch and FBD of the Pressure Vessel

7.4 Structural Analysis and Failure Criteria

First, the factor of safety is applied to the yield strength of the material, giving us a natural buffer to help ensure that the material does not fail.

$$\sigma_{allow} = \frac{\sigma_y}{FoS} \tag{7.1}$$

An inner radius is then chosen. This radius must be larger than the rod radius, and the piston seal on the rod will have the same radius.

$$A = \pi r_i^2 \tag{7.2}$$

$$P = \frac{F}{A} \tag{7.3}$$

After finding the area, the inner pressure is calculated to find out how much pressure is needed to push the maximum load F. No factor of safety is applied to this calculation because the pressure needs to be almost exact in order to safely translate the pulley bracket.

$$r_{o_{min}} = \sqrt{r_i^2 \frac{\sigma_{allow} + P}{\sigma_{allow} - P}} \tag{7.4}$$

The minimum outer radius $r_{o_{min}}$ is then calculated while keeping in the allowable stress, which leads to finding the minimum thickness to support the allowable stress.

$$t_{min} = r_{o_{min}} - r_i \tag{7.5}$$

Then, by rounding up the thickness $t > t_{min}$, it is ensured that the vessel will be able to withstand at least the allowable stress.

$$\frac{r_i}{t} \geq 10 \tag{7.6}$$

With the new, rounded up thickness, it can be determined whether the vessel is thin or thick-walled. If equation 7.6 is true, then the pressure vessel is thin-walled.

$$r_m = r_i + \frac{t}{2} \tag{7.7}$$

The mean radius is then calculated (7.7) in order to be plugged into the two thin-walled stress equations.

$$\sigma_h = \frac{(P)(r_m)}{t} \tag{7.8}$$

$$\sigma_l = \frac{(P)(r_m)}{2t} \tag{7.9}$$

The hoop σ_h and longitudinal σ_l stresses are then calculated by equations 7.8 and 7.9 respectively. These stresses must be smaller than the allowable stress calculated in equation 7.1 in order to ensure that the pressure vessel will not fail.

$$\sigma_{allow} = \frac{60\ kpsi}{2} = 30,000\ psi \tag{7.1}$$

The inner radius r_i of 2 inches has been chosen. It fits the requirements stated above for the pressure vessel.

$$A = \pi(2)^2 = 12.57\ in^2 \tag{7.2}$$

$$P = \frac{25{,}000\ lbf}{12.57\ in^2} = 1{,}990\ psi \tag{7.3}$$

The pressure vessel must be capable of producing an inner pressure of $1{,}990\ psi$.

$$r_{o_{min}} = \sqrt{(2\ in)^2 \frac{30{,}000\ psi + 1{,}900\ psi}{30{,}000\ psi - 1{,}900\ psi}} = 2.137\ in \tag{7.4}$$

$$t_{min} = 2.137\ in - 2\ in = 0.137\ in \tag{7.5}$$

Rounding up, the new thickness is $t = 0.15\ in$.

$$\frac{2\ in}{0.15\ in} = 13.33 \geq 10 \tag{7.6}$$

The new thickness meets the requirements of a thin-walled vessel, allowing the stress equations 7.8 and 7.9 to be applied.

$$r_m = 2\ in + \frac{0.15\ in}{2} = 2.075\ in \tag{7.7}$$

$$\sigma_h = \frac{(1{,}990\ psi)(2.075\ in)}{0.15\ in} = 27{,}515\ psi \tag{7.8}$$

$$\sigma_l = \frac{(1{,}990\ psi)(2.075\ in)}{2(0.15\ in)} = 13{,}758\ psi \tag{7.9}$$

8. Column Base (w/ Bolted Connection)

8.1 Description and Requirements

The column base, and more importantly the bolted connections secure the gantry crane. In most marine applications, which this crane is designed for, the vertical columns will have wheels mounted to the bottom to allow movement once removing a boat from the water. They could also be mounted to rails. These connections therefore must be strong enough to withstand the weight of a fully loaded crane, but the potential for an uneven load, or even the swing of an improperly secured boat. In order to model this, similar to the crossbar, the load is assumed to be at a 45 degree angle. This will put extreme stress on the bolts, and when designed to resist these stresses, create an overall stronger crane. Finally, the bolts will be analyzed for how tight they must be to meet the specifications. There were four holes in the bottom of the base for attachment to rails or wheels, and these can be seen in Figure 8.1.

8.2 Assumptions

For the Column Base, it has been assumed that there are no limits to the dimensions of the base, meaning that the wheels or rails will be designed to connect to the column base. It is

also assumed that the bolts in tension are the only ones resisting the moment of the column. Finally, the bolts are assumed to be lubricated which is normal when dealing with large industrial machinery.

8.3 Sketch and Free-Body Diagram

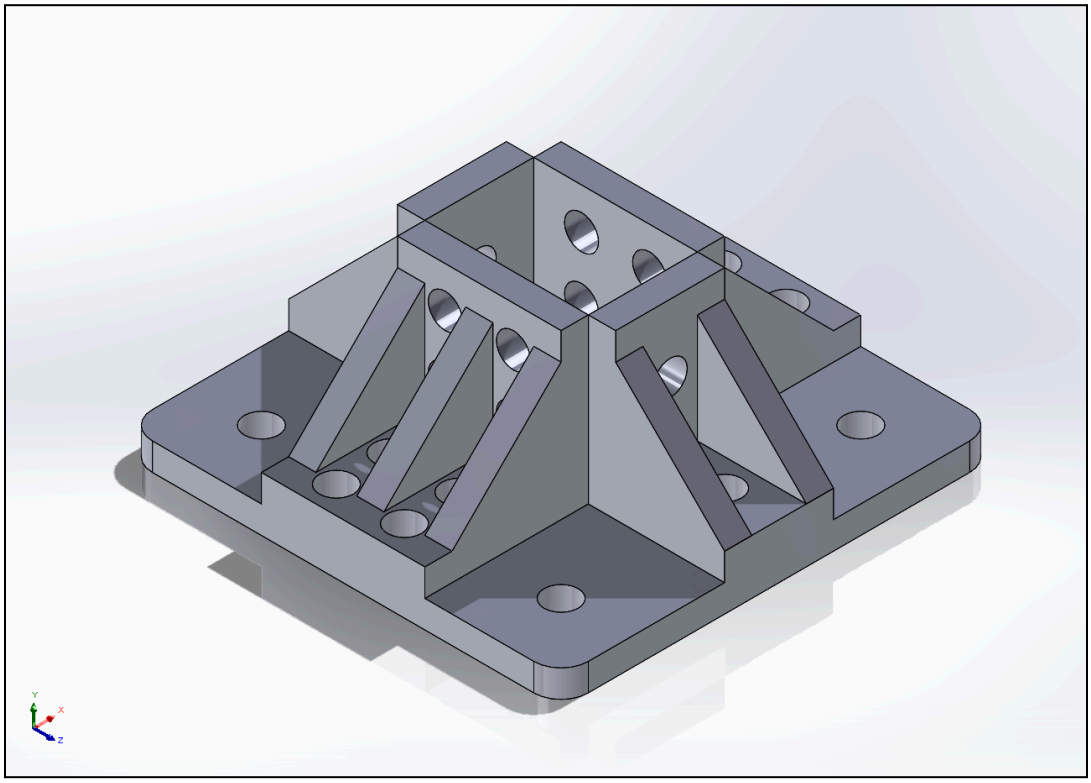


Figure 8.1: SolidWorks model of Column Base (no bolts)

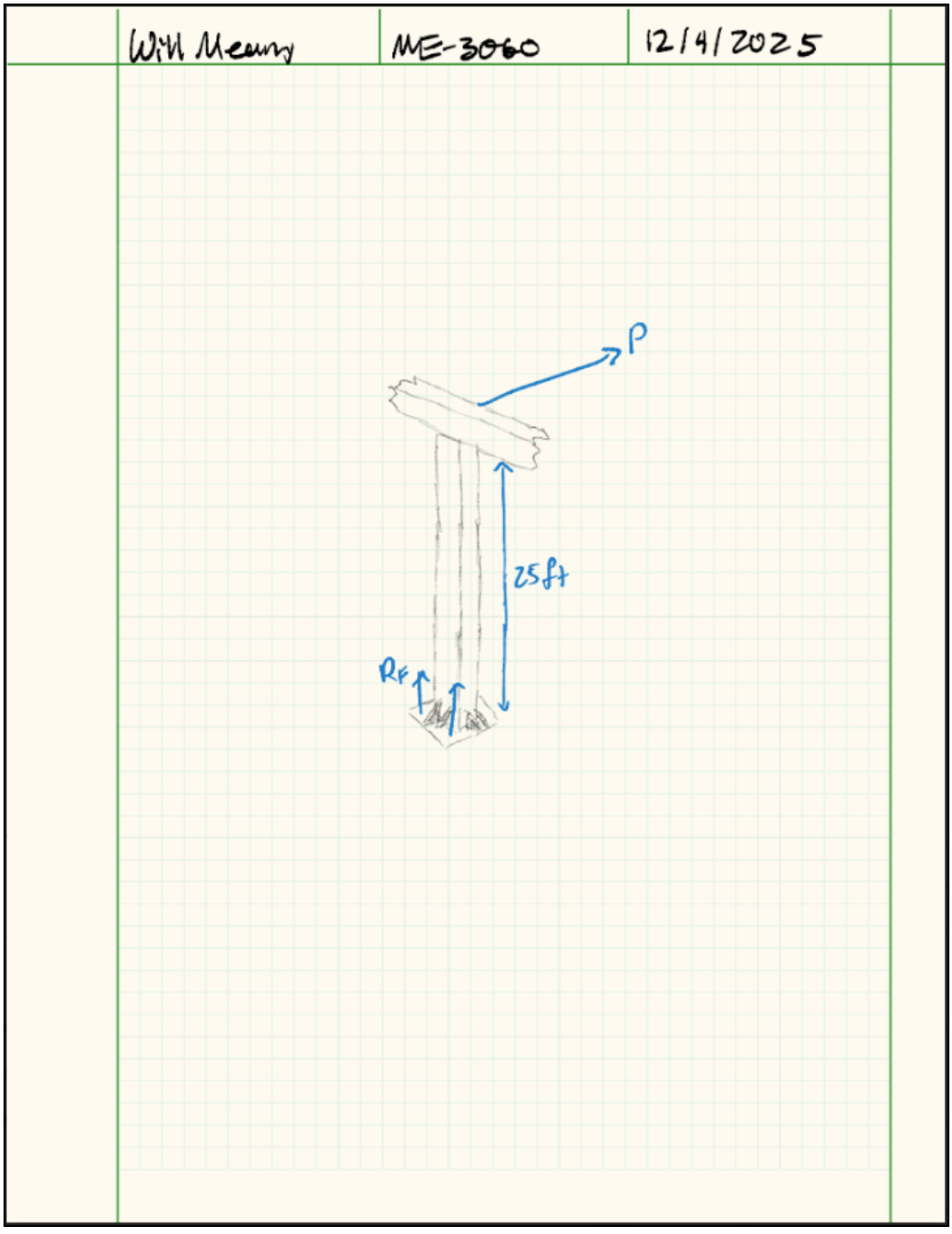


Figure 8.2: Free Body Diagram

8.4 Structural Analysis and Failure Criteria

The maximum shear will occur when the column is entirely bearing the force of a boat. The moment on the column base is extremely high, therefore the design involved using the biggest bolts available in the textbook. These bolts are SAE grade 8, 1-1/2in diameter. The failure point of the column base is in the bolts not being able to maintain tension, allowing the

gussets to separate from the column. Therefore, the analysis of the bolts is extremely important in ensuring that the crane meets or exceeds the factor of safety. The process begins with calculating the moment on the column base by the load.

$$M = P \cdot h \tag{8.1}$$

Next, the yield strength of the bolt is used to calculate the allowable stress.

$$\sigma_{allow} = \frac{\sigma_y}{n} \tag{8.2}$$

The area the tension acts over is given with the bolt characteristics, and is used to determine the allowable tension.

$$T_{allow} = \frac{\sigma_{allow}}{A} \tag{8.3}$$

Now that the allowable tension has been found, the tension on each bolt will be calculated using the number of bolts and the length of the bolts from the column.

$$T_{bolt} = \frac{M}{n_{bolts} \cdot l} \tag{8.4}$$

The tension in the bolts must be less than the allowable tension, and if it is not, more bolts can be added, or moved further away. The tightness of bolts must be calculated for proper assembly, and to ensure that they meet the design specifications. This process begins with the preload, which is found using the proof strength of the bolt, as well as the tensile stress area. This can be found in the tables in the textbook.

$$F_i = 0.75 \cdot S_p \cdot A_t \tag{8.5}$$

The final step to solve for tightness assumes that the bolt is lubricated, resulting in a k value of 0.18.

$$T = K \cdot D \cdot F_i \tag{8.6}$$

The resulting factor of safety can be found using the following equation.

$$n = \frac{T_{allow}}{T_{bolt}} \tag{8.7}$$

Plugging in values for SAE grade 8, 1-½ bolt,

$$M = 25,000lbf \cdot 25ft \cdot \frac{12in}{1ft} = 3,750,000lbf - in \tag{8.1}$$

$$\sigma_{allow} = \frac{130{,}000psi}{2} = 65,000psi \tag{8.2}$$

$$T_{allow} = \frac{65{,}000psi}{1.581in^2} = 102765lbf \tag{8.3}$$

$$T_{bolt} = \frac{3{,}750{,}000lbfin}{8 \cdot 10in} = 46875lbf \quad (8.4)$$

$$F_i = 0.75 \cdot 120{,}000psi \cdot 1.41in^2 = 126{,}900lbf \quad (8.5)$$

$$T = 0.18 \cdot 1.5in \cdot 126900lbf \cdot \frac{1ft}{12in} = 2855.25lbfft \quad (8.6)$$

$$n = \frac{102765lbf}{46875lbf} = 2.19 \quad (8.7)$$

The force requires 8 bolts to secure the column in place, and they must be tightened to a torque of 2855.25lbf-ft. These 8 bolts are all acting in tension, and more than meet the factor of safety due to there being more bolts securing the column base from other directions.

8.5 Finite Element Analysis

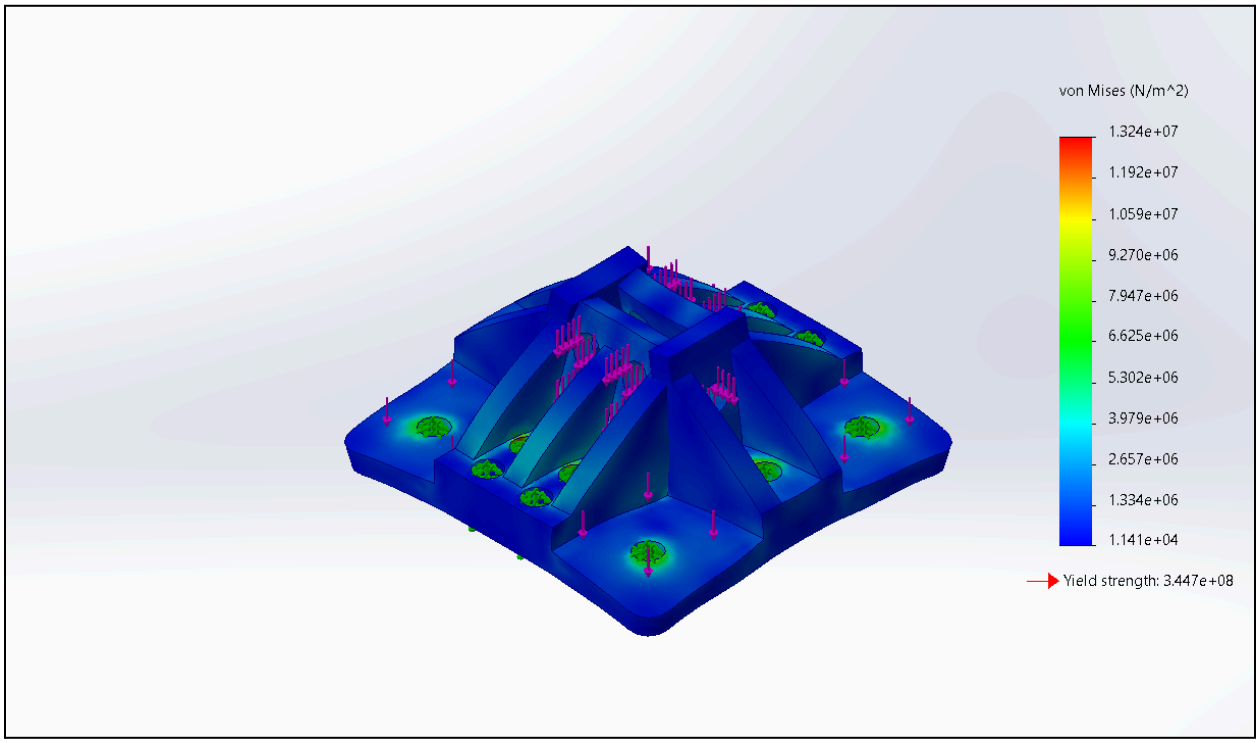


Figure 8.3: FEA of Column Base

Figure 8.3 shows the Finite Element Analysis of a singular column base. Here, the force from the weight of the boat will cause high stress along the holes where the bolts will be threaded into place.

9. Piston

9.1 Description and Requirements

The rod/piston is designed to be the linear extension that translates the pulley bracket forward and backward on the horizontal beams. The rod must be long enough to be able to cover all assumed-possible locations of lift of the pulley, where the center of mass of the boat is located.

The rod also interfaces with two separate components, the pressure vessel and the pulley bracket. The rod uses the pressure from the vessel in order to translate the bracket.

When designing the rod, both buckling and compression failure need to be taken into account to ensure that the system does not fail. Although both types of failures must consider both the shape and material of the part, buckling is the deformation/failure of the part (in this case the rod) due mostly to the geometry of the component. In this case, choosing the right length and diameter of the rod are vital to combat buckling. Conversely, compression failure mostly occurs due to the material strength and its inability to withstand stress, making our material selection very important.

Along with all these considerations, the rod must also be able to translate 12.5 tons (half of 25 tons). With these requirements, the rod was designed with a length of 10 ft, diameter of 2.75 in, and made from 4140 Quenched and Tempered steel ($\sigma_y = 60\ kpsi$). On each end of this rod, there are cylinders of length 2 in and diameter 4 in used to interface with its companion components.

With this design, illustrated below in figure 9.1, the rod is capable of effectively and safely moving the maximum load without failure (calculations provided below). It is also long enough to be able to cover the assumed range of where the pulleys need to be.

9.2 Assumptions

The pulleys need to be located within the same plane as the center of mass of the boat in order to lift only vertically. The rod needs to be long enough to cover the assumed range of where the center of mass will be.

9.3 Sketch and Free-Body Diagram

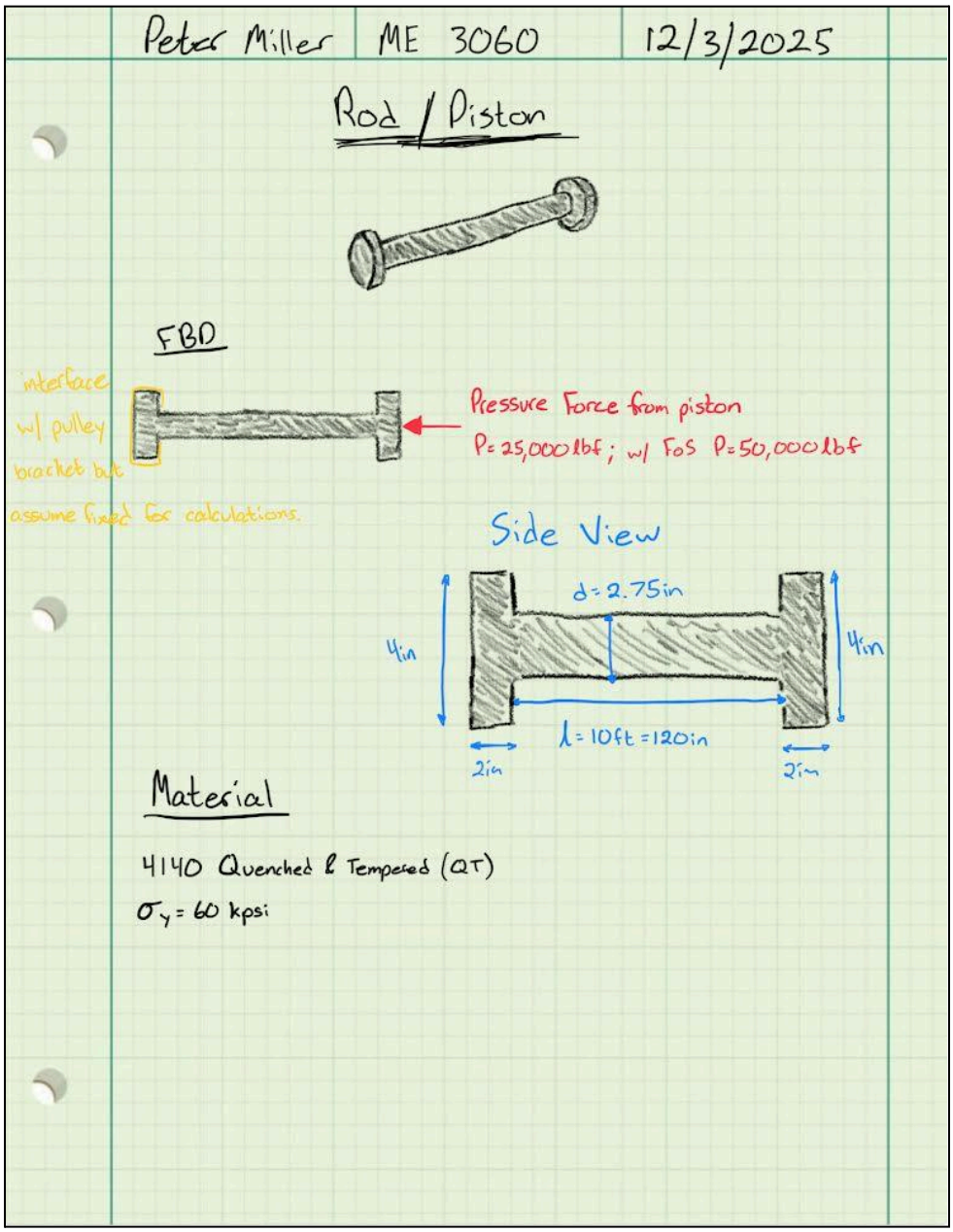


Figure 9.1: Sketch and FBD of the Rod/Piston

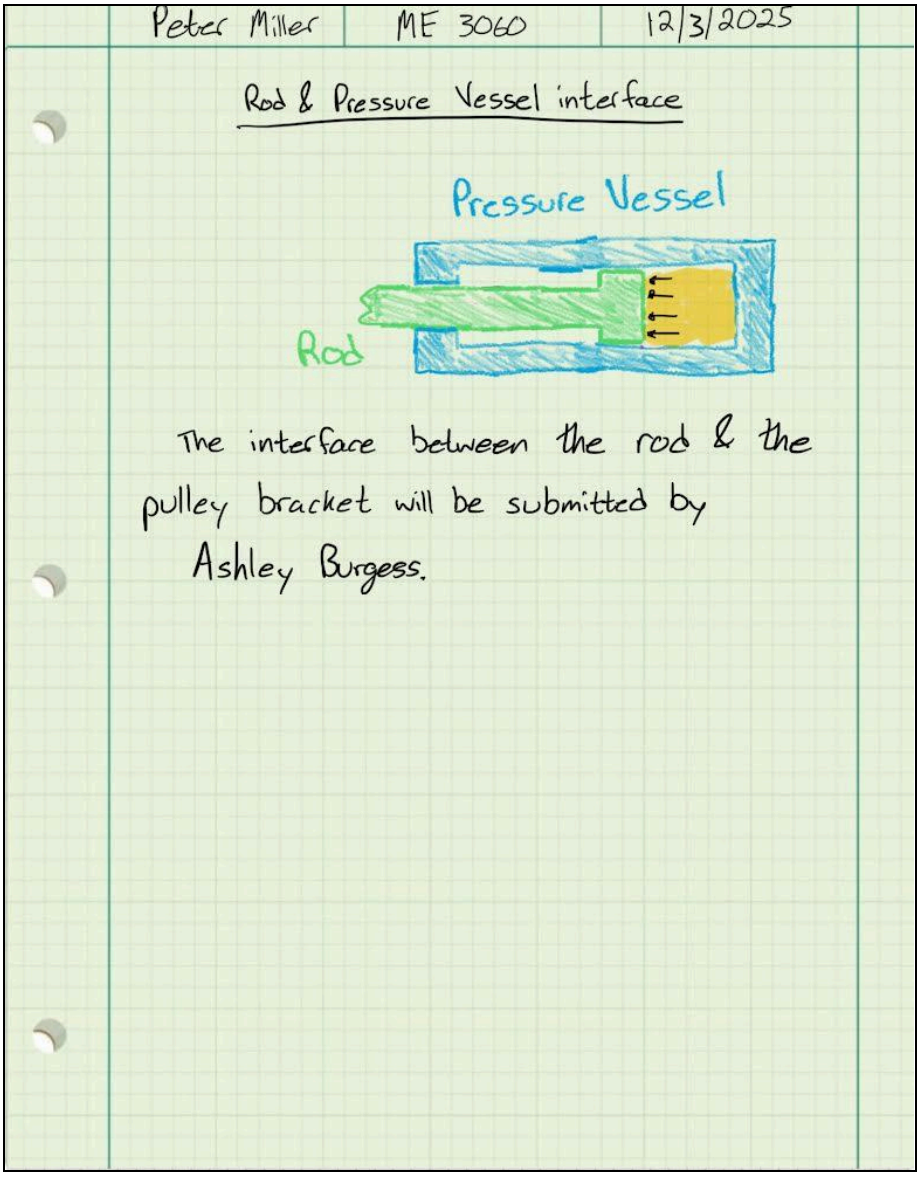


Figure 9.2: Sketch of the Interface Between the Rod and Pressure Vessel

9.4 Column Analysis

Applying the factor of safety to the maximum load, the calculations will have a built-in buffer to ensure the rod does not fail. The allowable load is assumed to be the same as the critical load, so when the diameter of the rod is later rounded up, the actual critical load the rod will be able to handle will be larger than the allowable load.

$$P_{allow} = P_{cr} = (P_{max})(FoS) \quad (9.1)$$

Then, the Euler Buckling equation used to calculate the critical load (9.2) and the Moment of Inertia equation of a rod (9.3) are combined to find the minimum diameter (9.4) required to withstand the allowable load.

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2} \quad (9.2)$$

$$I = \frac{\pi d^4}{64} \quad (9.3)$$

$$d_{min} = \left(\frac{64 P_{allow} (KL)^2}{\pi^3 E}\right)^{1/4} \quad (9.4)$$

The diameter is then rounded up to a standard manufacturing size, which ensures that the rod will not succumb to buckling because $P_{cr} > P_{allow}$

$$A = \frac{\pi d^2}{4} \quad (9.5)$$

$$\sigma_{comp} = \frac{P_{cr}}{A} \quad (9.6)$$

$$\sigma_{allow} = \frac{\sigma_y}{FoS} \quad (9.7)$$

To check the compression failure, the new (rounded) diameter is used to calculate the area (9.5) and the actual critical load (9.2). The actual stress from compression that the rod will sustain (9.6) must be less than the allowable stress found using the yield strength of the material in use and the factor of safety.

$$P_{cr} = (25,000\ lbf)(2) = 50,000\ lbf \quad (9.1)$$

$$E = 29\ MPa$$

$$L = 120\ in$$

$$K = 1.0$$

$$d_{min} = \left[\frac{(64)(50{,}000\ lbf)[(1)(120\ in)]^2}{\pi^3 (29\ MPa)}\right]^{1/4} = 2.676\ in \quad (9.4)$$

The diameter is rounded to $2.75\ in$, the next highest standard rod diameter.

$$A = \frac{\pi(2.75\ in)^2}{4} = 5.94\ in^2 \quad (9.5)$$

$$\sigma_{comp} = \frac{50{,}000\ lbf}{5.94\ in^2} = 8,419\ psi \quad (9.6)$$

The material chosen, 4140 Quenched and Tempered Steel, has a yield strength of $\sigma_y = 60\ kpsi$.

$$\sigma_{allow} = \frac{60{,}000\ psi}{2} = 30,000\ psi \quad (9.7)$$

The rod will not experience compression failure because $\sigma_{comp} < \sigma_{allow}$

10. Pulley

10.1 Description and Requirements

The pulley is designed to accept up to a ⅜" diameter steel cable. To standardize the design and ensure accessibility of replacement materials, the internal assembly is compatible with the McMaster-Carr standard corrosion-resistant flanged screw-clamp bushing. The mounting flange of the bushing prevents movement during tightening. [7] <INSERT BEARING> via an interference fit. Further, to ensure a solid construction, the bushing will be installed with a forced fit, which is the classification with the largest . [8] Given the pulleys' unique requirements and need to be machineable, it will be made out of AISI 4140 alloy steel. AISI 4140 steel has a allowable contact stress number s_{ac} (σ_{Hlim}) of 145000 lbf/in^2 and a normalized yield strength of 95 kpsi,

The pulley will move up and down the side bars via the pulley bracket, which is controlled by the piston rod. There are two pulleys on either side of the load, one on each of the horizontal side bars.

[7] https://www.mcmaster.com/products/pulley-bushings/pulley-and-sprocket-bushings-1~/corrosion-resistant-flanged-screw-clamp-bushings/

[8] https://fractory.com/limits-and-fits/

[9] https://www.ckit.co.za/secure/conveyor/papers/troughed/modern/modern.htm

10.2 Assumptions

10.3 Sketch and Free-Body Diagram

10.4 Structural Analysis and Failure Criteria

To begin, the maximum and minimum radial interference of the pulley must be calculated using equation 10.1 and 10.2 below.

$$\delta_{max} = 0.5(d_{max\ shaft} - D_{min\ hole}) \quad (10.1)$$

$$\delta_{min} = 0.5(d_{min\ shaft} - D_{max\ hole}) \quad (10.2)$$

Where the lower case 'd' is the diameter of the shaft, and upper case 'D' is is the diameter of the hole, according to notation convention, and δ is the radial interference. Since this is a local interference fit classification, the maximum and minimum interference pressures must be calculated using the equations below.

$$Pmax = (E\,\delta_{max})/(2 * R)\,[1 - (R/r_0)^2] \quad (10.3)$$

$$Pmin = (E\,\delta_{min})/(2 * R)\,[1 - (R/r_0)^2] \quad (10.4)$$

Where ‘P’ is the interference pressure, ‘E’ is the modulus of elasticity of the metal, in this case given to be _____, ‘R’ is the interference radius,

11. Crane Hook

11.1 Description and Requirements

The crane hook is a crucial component in the crane. It is how the body of the crane will interact with the boats it is designed to lift. The hook in this crane is designed for 2in diameter cable, but other designs will use many straps to evenly spread the load of the lift about the width and length of the hull. Due to this important role, the crane hook will have its material chosen independently of other components on the crane. This hook is designed to take half of the boat's weight, and therefore must be designed for 12.5 tons. A large factor of safety will be implemented here due to the importance of its role, and potential failure creating a domino effect and causing other components to fail. A failure, which could be shear, bending, or any other way the cable is no longer secured could cause damage to the crane, as well as the boat being lifted. It is for this reason that the factor of safety was decided to be 2.5.

11.2 Assumptions

While designing the crane hook a few assumptions were made during the design process. Assumption one is that the load is static, and there are no sudden jumps in forces. This means that the crane should lift slowly such that there is no jolt. Considering the large factor of safety, this could still pose an issue. Third, is that the cross section is uniform, and follows curved beam theory. Finally, when designing a curved beam, increasing the height results in diminishing returns in strength. This is combated by increasing the width. This results in a very wide crane hook, but gantry cranes normally have multiple hooks spaced out in order to deal with this large hook issue.

11.3 Sketch and Free-Body Diagram

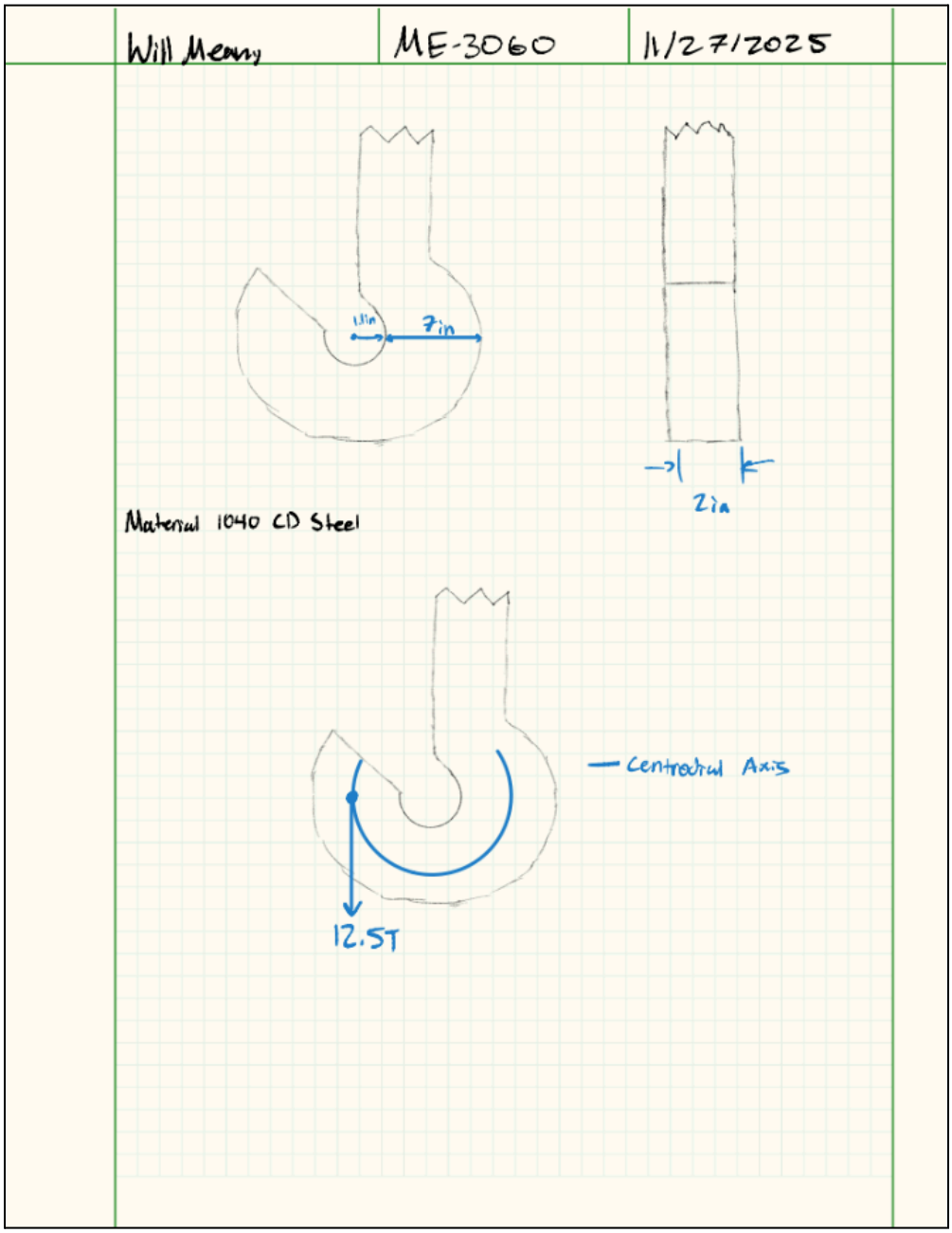


Figure 11.3: Crane Hook Free-body Diagram and Sketch

11.4 Structural Analysis and Failure Criteria

The crane hook was designed by identifying the dependent variables. These are the variables that are dependent on other variables, or are defined by the problem statement or constraints. The remaining variables were the independent variables, and are as follows: inside radius, height, width. Inside radius had to be greater than 1 inch, and the remaining height and

width could be adjusted to meet the required strength. After plugging in many values, the height of 10 inches, and width of 8.5 inches was a perfect fit. This analysis will use a few equations listed below. r_c is the radius of the centroid, r_i is the inside radius, and finally h is the height of the member. Centrodial radius is where the moment will be taken.

$$r_c = r_i + \frac{h}{2} \tag{11.1}$$

Once the centroidal radius is found, the neutral axis r_n could be found below.

$$r_n = \frac{h}{ln\frac{r_c}{r_i}} \tag{11.2}$$

The next equation needed is for e, which is simply found by subtracting results from the previous equations.

$$e = r_c - r_n \tag{11.3}$$

Next, the distances from the centroidal axis to the inside and outside fibers of the beam are calculated.

$$c_i = r_n - r_i \tag{11.4}$$

$$c_o = r_o - r_n \tag{11.5}$$

Finally, the necessary variables for the stress on the inside and outside fibers have been solved for, and can now be plugged in. A is the area, and M is the moment taken about the centroidal axis.

$$\sigma_i = \frac{Mc_i}{Aer_i} \tag{11.6}$$

$$\sigma_i = \frac{Mc_o}{Aer_o} \tag{11.7}$$

Finally, the factor of safety must be considered along the maximum stress, this is done with the next equation where n represents the factor of safety, and S_y is the yield strength.

$$n = \frac{S_y}{\sigma_{max}} \tag{11.8}$$

Using r_i = 1.1in, h = 7in, w = 2 in

$$r_c = 1.1in + \frac{12in}{2} = 4.6in \tag{11.1}$$

$$r_n = \frac{10in}{ln\frac{7.1in}{1.1in}} = 3.51in \tag{11.2}$$

$$e = 7.1in - 4.84in = 1.09in \tag{11.3}$$

$$c_i = 4.84in - 1.1in = 2.41in \tag{11.4}$$

$$c_o = 13.1in - 4.84in = 4.59in \tag{11.5}$$

$$\sigma_i = \frac{12.5ton \cdot \frac{2000lb}{ton} \cdot 4.6in \cdot 2.41in \cdot \frac{1kpsi}{1000psi}}{12in^2 \cdot 1.09in \cdot 1.1in} = 16.42kpsi \tag{11.6}$$

$$\sigma_o = \frac{12.5ton \cdot \frac{2000lb}{ton} \cdot 4.6in \cdot 4.59in \cdot \frac{1kpsi}{1000psi}}{12in^2 \cdot 1.09in \cdot 8.1in} = 4.26kpsi \tag{11.7}$$

$$n = \frac{84kpsi}{16.42kpsi} = 5.11 \tag{11.8}$$

With a factor of safety of 5.11, this hook design more than exceeds what is required. This is due to the extreme importance of this part, and extraneous causes of stress. If lifting a boat, the cable might snap the hook tight causing an instantaneous increase in the stress. These are the reasons careful consideration must be taken when designing the crane hook. It is for this importance that fatigue analysis will be performed on the hook as well.

11.5 Fatigue Analysis

The fatigue analysis will occur on the inside of the hook. This is where the maximum stress is, and therefore most likely to experience yielding over time. Fatigue analysis begins with solving the alternating and mean stresses. The σ_{min} is zero due to there being no forces on the crane hook when unloaded.

$$\sigma_a = \frac{\sigma_{max} - \sigma_{min}}{2} \tag{11.9}$$

$$\sigma_m = \frac{\sigma_{max} + \sigma_{min}}{2} \tag{11.10}$$

Once these have been solved for, finding the endurance strength (S_e) is the next step. This is done in a series of smaller steps which analyze the surface, size, load, temperature, and reliability.

$$S_e = k_a \cdot k_b \cdot k_c \cdot k_d \cdot k_e \cdot S_e' \tag{11.11}$$

$$k_a = a \cdot S_{ut}^{\ b} \tag{11.12}$$

$$k_b = 0.91d^{-0.157} \tag{11.13}$$

$$k_c = 1 \tag{11.14}$$

$$k_d = temperature\ factor \tag{11.15}$$

$$k_e = reliability\ factor \tag{11.16}$$

$$S_e' = 0.5S_{ut} \tag{11.17}$$

The k_d value is the temperature factor, and will not be significant due to the crane operating at normal temperatures. The k_e value is the reliability factor, which was chosen to be 99%. This returns a value of 0.814. With the endurance strength solved for, the values are plugged into a modified Goodman equation to determine if the hook is expected to experience infinite life.

$$n_f = \left(\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}}\right)^{-1} \tag{11.18}$$

If the Goodman equation returns a value greater than one, then infinite life is predicted. Plugging in results from stress analysis of the crane hook,

$$\sigma_a = \frac{16.42kpsi - 0}{2} = 8.21kpsi \tag{11.9}$$

$$\sigma_m = \frac{16.42kpsi + 0}{2} = 8.21kpsi \tag{11.10}$$

$$k_a = 2.7 \cdot 85^{-0.265} = 0.832 \tag{11.12}$$

$$k_b = 0.91(7.1)^{-0.157} = 0.669 \tag{11.13}$$

$$k_c = 1 \tag{11.14}$$

$$k_d = 1 \tag{11.15}$$

$$k_e = 0.814 \tag{11.16}$$

$$S_e' = 0.5(85kpsi) = 42.5kpsi \tag{11.17}$$

$$S_e = 0.832 \cdot 0.669 \cdot 1 \cdot 1 \cdot 0.814 \cdot 42.5kpsi = 19.26kpsi \tag{11.11}$$

$$n_f = \left(\frac{8.21kpsi}{19.26kpsi} + \frac{8.21kpsi}{85kpsi}\right)^{-1} = 1.91 \tag{11.18}$$

Therefore infinite life of the hook is predicted due to $n_f > 1$. This is due to the large factor of safety incorporated into the crane design, and will potentially save the client money due to not having to replace the crane hook. This also reduces the probability of the hook giving out after extended regular usage.

12. Conclusion

Designing a gantry crane is difficult. There are countless variables, and it is impossible to manage them all up front. The process was iterative, and work had to be changed frequently based on other sections being completed to ensure the assembly would come together. The method used for this project involved breaking down the crane into its individual components and working out from there. This had a great advantage in which multiple people were able to progress at once, but occasionally there would be overlap, and calculations would have to be repeated. The usage of spreadsheets allowed for quick changes to calculations, and for multiple independent variables to be tested in a short period of time. This almost eliminated the downsides of the individual component method.

In terms of limitations to the design, these come through the assumptions. A few major assumptions were made in order to keep the calculations similar to what was discussed in class. The factor of safety allows for leeway, and the design could still be applied or produced. In terms of physical limitations, the 25ft height of the crane is tall, and would get taller with wheels, it can be a limitation. The biggest limitation is the length. Due to the restriction of the hydraulic cylinders having to be as long as their pistons, there was not a reason to extend our crane beyond its current dimensions.

This has been a big project, and has allowed some big takeaways. The engineering process has been taught, but this represents the first occasion to apply it to a project that feels real. Working and proofreading has forced me to return to the previous content from the course and re-learn topics. Next, holding ourselves to a professional standard has been tough. Every sentence must be essential, but someone who knows nothing about the topic should be able to understand this report. Finally, we are very proud of the work and effort that has been put into this design project, and hope to apply it to future projects.

13. References

"Boat Size Classifications & Requirements." *BOATERexam.com®*, www.boaterexam.com/boating-resources/boat-size-classifications/.

Budynas, Richard G, et al. *Shigley's Mechanical Engineering Design*. 11th ed., New York, Ny, Mcgraw-Hill Education, 2020.

EnduraSteelAdmin. "How to Choose the Right Steel Grade for Your Project." *Endura Steel*, 31 Mar. 2023, endura-steel.com/how-to-choose-the-right-steel-grade-for-your-project/.

https://www.facebook.com/dmitry.shafran. "Boat Weight Estimates for 7 Weight Boat Types." *Maritime Page*, 28 Aug. 2022, maritimepage.com/boat-weight-estimates-for-7-weight-boat-types/. Accessed 5 Dec. 2025.

"MetalsDepot® - Buy Rectangle Steel Tube Online!" *Www.metalsdepot.com*, www.metalsdepot.com/steel-products/steel-rectangle-tube.

treywaldman. "Understanding Steel Corrosion: Causes, Signs, and Impacts - Waldman Engineering Consultants, Inc." *Waldman Engineering Consultants, Inc.*, 5 Sept. 2024, waldmaneng.com/2024/09/05/understanding-steel-corrosion-causes-signs-and-impacts/.

wide-marketing. "Safety Factor for Lifting Devices." *Airpes*, 28 Apr. 2022, www.airpes.com/safety-factor-lifting-devices/.